



TEST CODE **02120020**

FORM TP 2016199

MAY/JUNE 2016

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

ENVIRONMENTAL SCIENCE

ECOLOGY, PEOPLE AND NATURAL RESOURCE USE

UNIT 1 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions, TWO from each Module.
2. Answer ALL questions.
3. Write your answers in the spaces provided in this booklet.
4. Do NOT write in the margins.
5. You may use a silent, non-programmable, scientific calculator to answer questions.
6. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
7. **If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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MODULE 1

Answer **BOTH** questions.

1. (a) Distinguish between sedimentary biogeochemical cycles and gaseous biogeochemical cycles. In your response give ONE example of a sedimentary biogeochemical cycle and ONE example of a gaseous biogeochemical cycle.

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[6 marks]

- (b) Figure 1 shows the results of an investigation of the carrying capacity in a forest area by two different students, Student A and Student B, for a particular species. Study Figure 1 and answer the questions that follow.

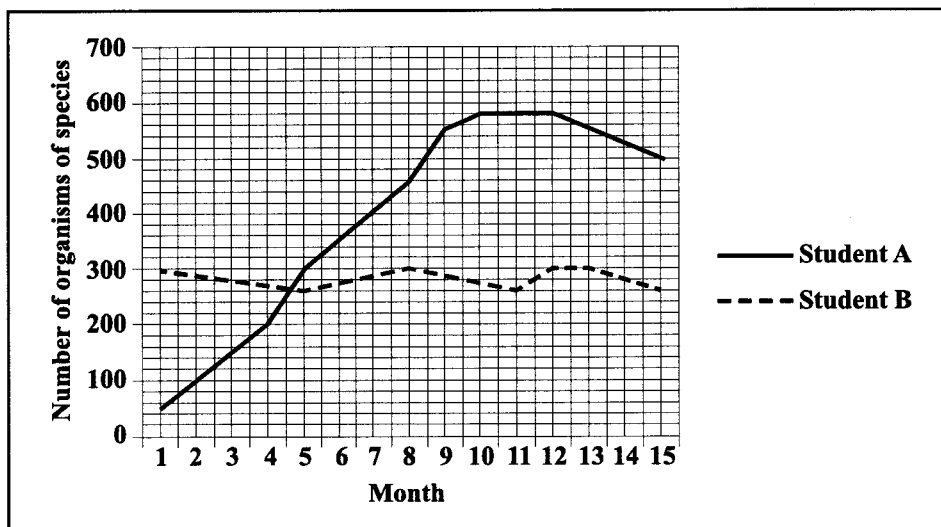


Figure 1. Results of the investigation of the carrying capacity in a forest area

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- (i) Make THREE deductions about the variation of the population of the species obtained by Student A.

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[3 marks]

- (ii) Calculate the difference in the carrying capacity values obtained by Student A and Student B.

[3 marks]

- (iii) Suggest TWO reasons for the difference in carrying capacities obtained by the two students.

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[4 marks]

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- [illegible]

Total 20 marks

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2. (a) State THREE ways in which primary ecological succession is different from secondary ecological succession.

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[3 marks]

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- (b) Figure 2 shows the percentage deforestation that occurred in three countries: A, B and C. Study Figure 2 and answer the questions that follow.

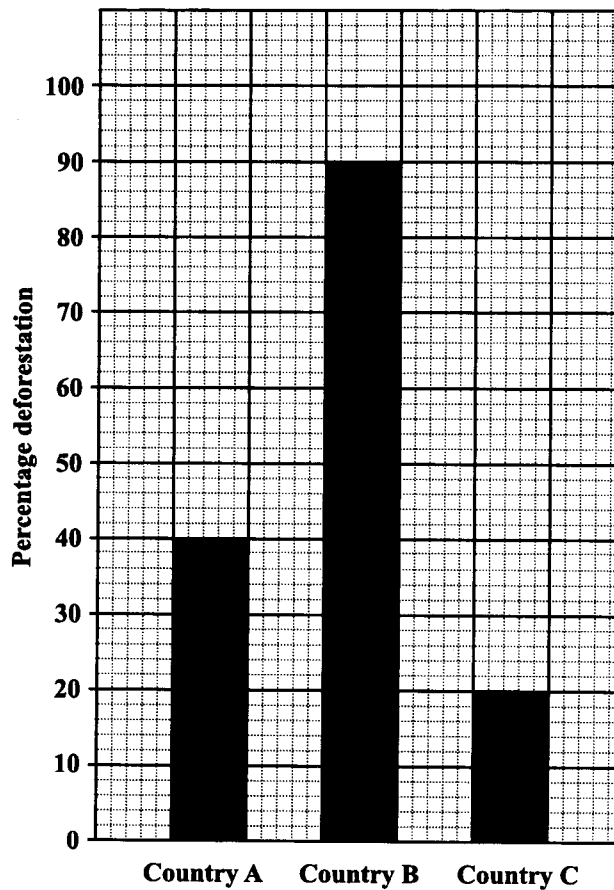


Figure 2. Percentage deforestation that occurred in Country A, Country B and Country C

- (i) Use Figure 2 to determine which two countries have a combined total of 70% of their forest intact. Show your working.

[4 marks]



- (ii) State the country in which ecological succession is MOST likely to occur and justify your response.

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[7 marks]

- (c) Define the term 'limiting factor' and explain how limiting factors operate in the environment.

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[3 marks]

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- (d) Explain the role of natural selection in the adaptation of species to their natural environment.

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[3 marks]

Total 20 marks



MODULE 2

Answer BOTH questions.

3. (a) Table 1 presents data for water consumption per capita for developed and developing countries.

**TABLE 1: WATER CONSUMPTION PER CAPITA FOR DEVELOPED
AND DEVELOPING COUNTRIES**

Type of Country	Water Consumption per Capita (m ³)
Developed	2840
Developing	1070

- (i) Calculate the percentage difference in the per capita water consumption for developing and developed countries.

[4 marks]

- (ii) Account for the difference in per capita water consumption for developing and developed countries.

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[4 marks]

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- (iii) Discuss the environmental impact which may occur as a result of high per capita water consumption.

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[6 marks]

- (b) Explain how EACH of the following may impact on the rate of growth of a population:

- (i) Religion

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[3 marks]

- (ii) Culture

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[3 marks]

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4. (a) What is meant by the term 'sustainable development'?

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[3 marks]

- (b) Explain how high birth rates may impact on a country's ability to achieve sustainable development.

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[4 marks]

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- (c) Table 2 shows the Human Development Index (HDI) for two countries, Country A and Country B.

**TABLE 2: HUMAN DEVELOPMENT INDEX (HDI)
FOR COUNTRY A AND COUNTRY B**

Country	HDI
A	0.449
B	0.955

Use the two values in the table to make TWO comparisons between Country A and Country B.

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[2 marks]

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- (d) Figure 3 shows the percentage of the world's population that is affected by the lack of access to selected social services. Study Figure 3 and answer the questions that follow. In this question assume that the world's total population is 7.1 billion.

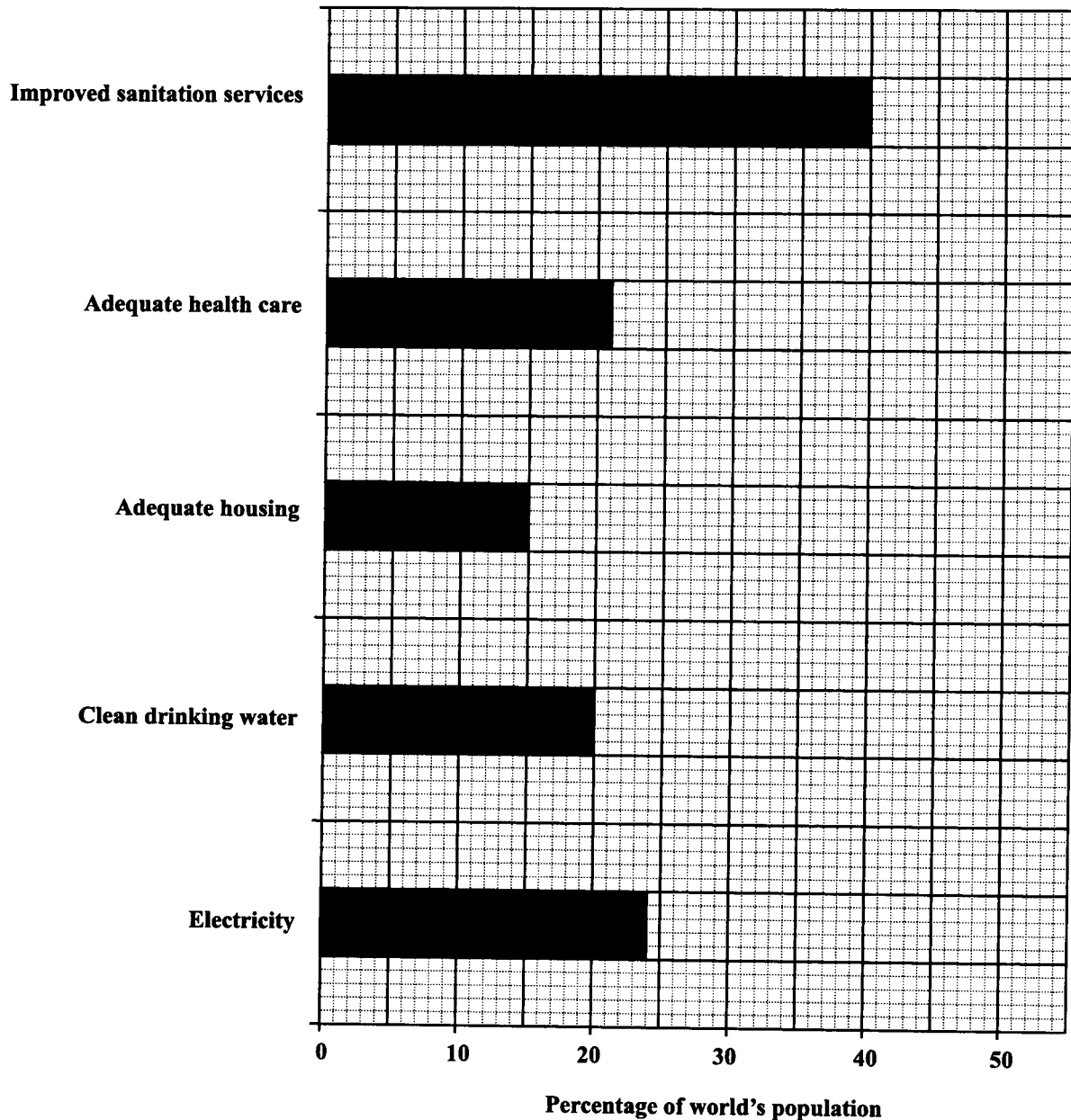


Figure 3. Percentage of the world's population affected by lack of access to selected social services

- (i) What percentage of the world's population lacks access to adequate health care services?

[1 mark]



- (ii) Calculate the total number of people who lack access to BOTH clean drinking water and improved sanitation services.

[4 marks]

- (iii) Poor sanitation services may lead to increased levels of pollution. Explain THREE different ways by which the impact of pollution may be mitigated.

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[6 marks]

Total 20 marks

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MODULE 3

Answer BOTH questions

5. Figure 4 is a diagram of coastal areas with cliffs, forests, hills, a river which runs into the ocean, some oil rigs and a quarry.

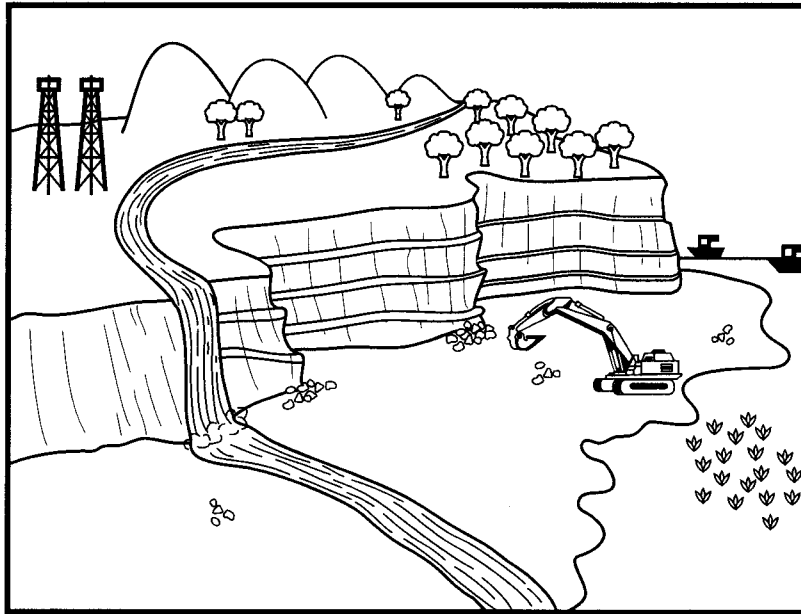


Figure 4. Diagram showing coastal areas

- (a) Identify TWO types of natural resources illustrated in Figure 4.

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[2 marks]

- (b) Choose ONE of the natural resources identified in (a) above and answer the following questions

- (i) Classify the selected natural resource as renewable or non-renewable and exhaustible or inexhaustible.

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[2 marks]



- (ii) Using a selected natural resource as an example, explain the difference between 'non-renewable natural resources' and 'exhaustible natural resources'.

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[3 marks]

- (c) You overhear a student saying that a natural resource cannot be important for both livelihood AND recreation. State whether this statement is true or false. Using ONE of the natural resources illustrated in Figure 4 as an example, justify your response.

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[6 marks]



- (d) Table 3 shows the Amerindian population as a percentage of the total population in selected Caribbean countries.

TABLE 3: AMERINDIAN POPULATION AS A PERCENTAGE OF THE TOTAL POPULATION IN SELECTED CARIBBEAN COUNTRIES

Country	Percentage of Amerindians in the Total Population (%)
Belize	15
Dominica	6
Guyana	9
St Vincent and the Grenadines	2
Suriname	2
Trinidad and Tobago	0.1

- (i) Sketch a bar chart illustrating the data presented in Table 3. A detailed graph is not required.

[3 marks]

- (ii) Given that the total population of Trinidad and Tobago is approximately 1.3 million, how many Amerindians are there?

[2 marks]

- (iii) Outline ONE practice that the indigenous peoples of the Caribbean use to manage their natural resources.

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[2 marks]

Total 20 marks

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6. (a) Define the term 'non-consumptive use of natural resources' and identify ONE example that illustrates non-consumptive use of natural resources.

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[2 marks]

- (b) (i) Three tools that can be used for natural resource management and conservation are sustainable yield management, waste minimization and environmental legislation. Choose ONE of these tools and explain how it works to achieve natural resource management and conservation.

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[4 marks]

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- (ii) Explain how political factors can influence the successful use of the tools for natural resource management.

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- (c) Figure 5 shows the number of threatened plant and animal species in selected Caribbean territories. Study Figure 5 carefully and answer the questions that follow.

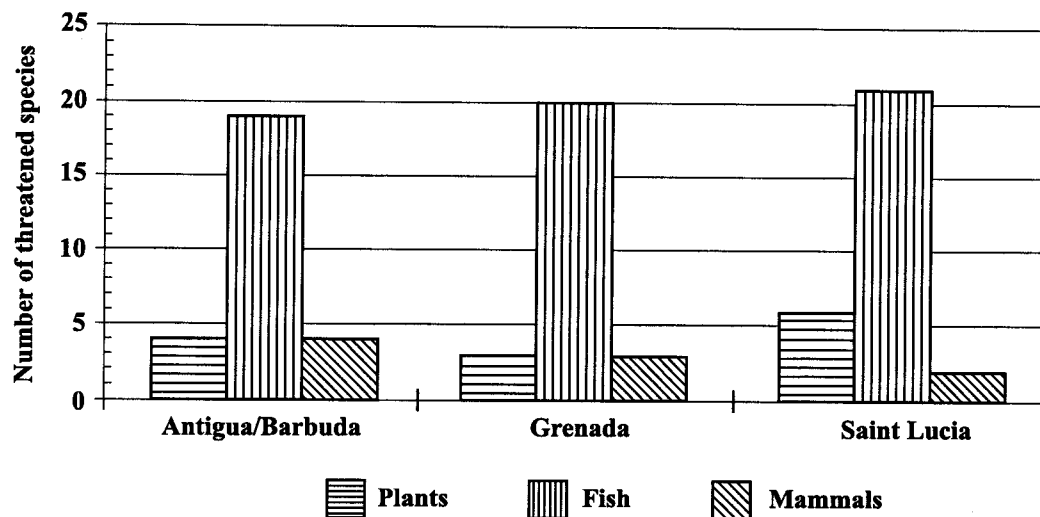


Figure 5. Graph showing the number of threatened plant and animal species in selected Caribbean territories

Source: The World Bank, Environmental Indicators, www.data.worldbank.org/indicator.

- (i) Using a suitable table, summarize the information shown in Figure 5.

[5 marks]

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- (ii) Suggest THREE reasons why fish are the most threatened species in all three territories.

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[6 marks]

Total 20 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.



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